

Deep Neural Networks - Mid-Semester 1 Exam Guide

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DNN Midsem 1 Preparation Hub

Course: Deep Neural Networks. Exam: 26 September 2026, closed book, 09:00 AM-11:00 AM. Scope: Contact Sessions 1-8.

Schedule note: a [user-provided schedule record](#) lists AMLCS ZG511 on 26 September 2026, 09:00 AM-11:00 AM. No original calendar URL is available for independent verification; the handout still says the date/time are “To be announced.”

Start with the [syllabus map and mini-lessons](#). Printable guide: [DNN Midsem 1 exam guide PDF](#).

Six-day plan

Day 1 - 20 September: perceptrons and MLPs

Learn [Sessions 1-2](#), read [Session 1](#) and [Session 2](#), then attempt [C01-C06](#) and [N01-N06](#).

Day 2 - 21 September: propagation and outputs

Learn [Session 3](#), read [Session 3](#), then attempt [C07-C09](#) and [N07-N09](#).

Day 3 - 22 September: optimization

Learn [Session 4](#), read [Session 4](#), then attempt [C10-C12](#) and [N10-N12](#).

Day 4 - 23 September: regularization and stability

Learn [Session 5](#), read [Session 5](#), then attempt [C13-C15](#) and [N13-N15](#).

Day 5 - 24 September: CNNs and transfer

Learn [Sessions 6-7](#), read [Session 6](#) and [Session 7](#), then attempt [C16-C21](#), [N16-N21](#), and the [source-aware practice](#).

Day 6 - 25 September: integrate, test, check, recall

Read [Session 8](#), attempt [C22-C24](#) and [N22-N24](#), then sit [Mock 01](#) for 90 minutes. Only after submission, use the [answer key](#), [rubric](#), or [solutions PDF](#). Finish with the [last-30-minutes route](#).

For every calculation-bearing answer, show formula, substitution, intermediate steps, units/shapes/conditions, and interpretation.

DNN Midsem 1 Syllabus Map and Mini-Lessons

The handout prescribes Sessions 1-8. “Local evidence” only reports matching repository artifacts; it does not alter prescribed scope.

Session	Handout-prescribed scope	Local teaching evidence
1	Learning paradigms; motivation/applications; biological analogy; connectionism; perceptron, learning, XOR	Strong

Session	Handout-prescribed scope	Local teaching evidence
2	Perceptron; MLP classifiers and universal approximators; depth and width	Strong
3	Forward/backpropagation; computation graphs; GD; activations; softmax; depth	Introductory only
4	Nonconvex challenges; SGD/minibatches; Rprop, Quickprop, Nesterov, Adagrad, RMSprop, Adam	Not substantive
5	Model selection; under/overfit; L1/L2; dropout; gradients; initialization; covariance shift; BN	Not substantive
6	Image convolution; learned kernels; padding, stride, channels, pooling, CNN design	Not substantive
7	CNN architectures; transfer learning; applications	Not substantive
8	Review of Sessions 1-7	No separate local review artifact

Prerequisite bridge

A scalar is one number; a vector is an ordered column; a matrix is a rectangular array. An $m \times n$ matrix maps an $n \times 1$ input to an $m \times 1$ output. Superscript T means transpose: turn a column into a row, so $w^T x$ is the dot product $w_1 x_1 + \dots + w_d x_d$. An affine map is a linear matrix multiplication plus a constant bias. A derivative measures local change; a gradient collects a scalar output's derivatives with respect to several inputs. A probability lies from 0 to 1. $\exp(a)$ means the exponential e raised to a ; $\log$ is its inverse. $\text{floor}(r)$ means round r down to the nearest integer. "Elementwise" means apply the operation separately to matching entries.

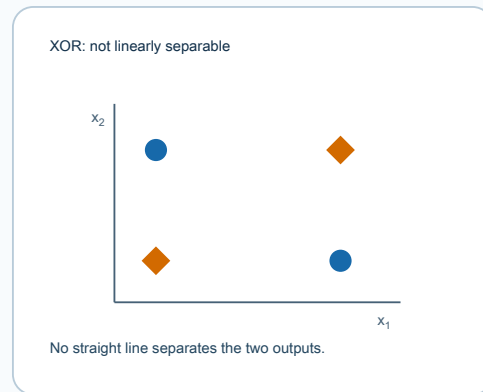
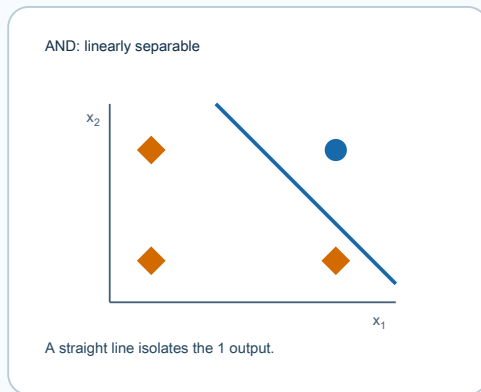
Sessions 1-2 - Fundamentals and multilayer perceptrons (MLPs)

An artificial neuron computes $z = w^T x + b$, then $\hat{y} = f(z)$. Here x is the feature vector, w the weight vector, b the bias, z the score, f the activation, and $\hat{y}$ the prediction. Under the fixed step convention, $z \geq 0$ maps to 1. In the perceptron update $w \leftarrow w + \eta(y - \hat{y})x$, $b \leftarrow b + \eta(y - \hat{y})$, y is the correct label and the learning rate $\eta > 0$ controls step size. Data are linearly separable when one line (or flat higher-dimensional boundary) separates the two classes. The finite-convergence guarantee needs a finite linearly separable dataset and the standard repeated-update procedure.

For AND with $w = (1, 1)$, exact boundary reasoning gives $-2 \leq b < -1$. XOR is not linearly separable, so a single perceptron cannot solve it. A nonlinear MLP can create intermediate features and combine boundaries.

AND vs. XOR

A visual reason one linear perceptron cannot solve XOR



single perceptron

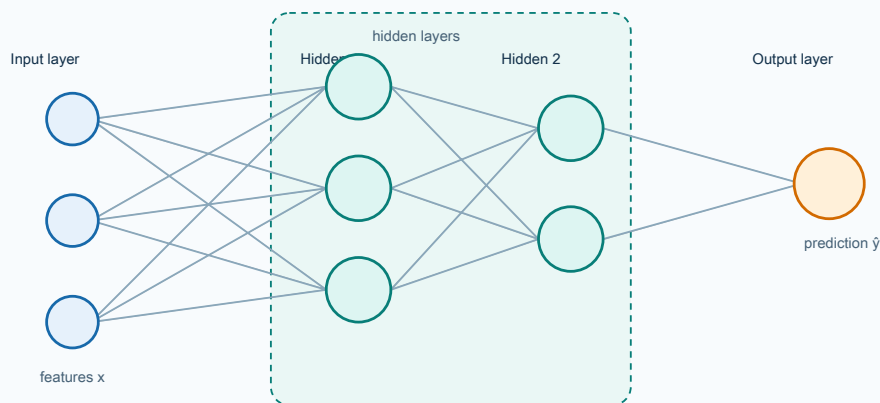
layered network can represent it

● output 1 ◆ output 0

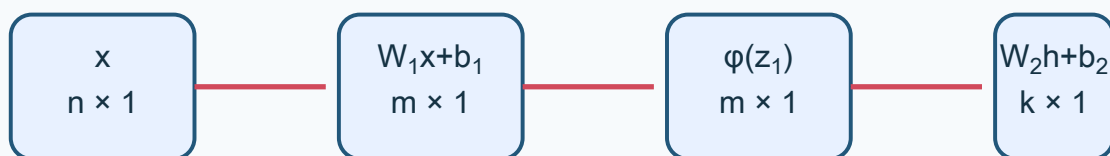
A dense layer uses $z=Wx+b$; if W is $m \times n$, its bias and output are $m \times 1$, and it has $mn+m$ parameters. Width counts units; depth counts computational transformations under a stated convention. Universal approximation is an existence/capacity result under stated activation/domain conditions—not a training, efficiency, or generalization guarantee.

A neural network is organized in layers

An MLP connects an input layer, hidden layer(s), and an output layer



More than one hidden layer is the course's intuitive meaning of a deep neural network.

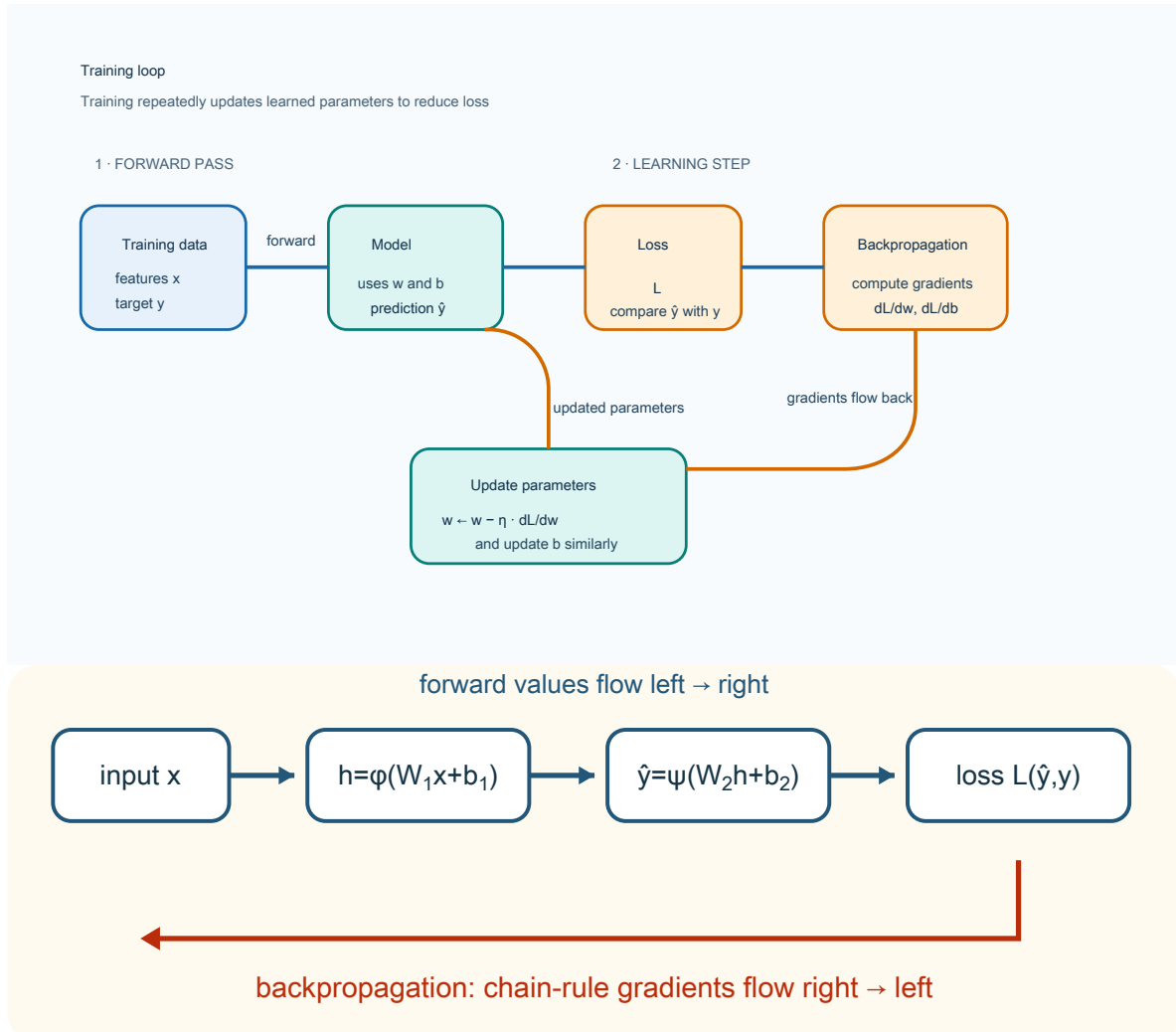


$W_1: m \times n, b_1: m \times 1; W_2: k \times m, b_2: k \times 1$

Worked example. A $4 \rightarrow 3 \rightarrow 2$ network has $4 \cdot 3 + 3 + 3 \cdot 2 + 2 = 23$ parameters. The input placeholder is not counted as a trainable layer in this statement.

Session 3 - Deep feedforward neural networks

A feedforward network repeatedly applies affine maps and activations. The forward pass computes activations and loss. A computation graph records which value depends on which earlier value. The chain rule multiplies local derivatives along such dependencies. Backpropagation applies it backward to calculate parameter gradients; gradient descent moves each parameter opposite its loss gradient.



A logit is a class score before probability conversion. Sigmoid is $1/(1+\exp(-z))$ and maps one logit to $(0,1)$; tanh is $(\exp(z)-\exp(-z))/(\exp(z)+\exp(-z))$ and maps to $(-1,1)$. Both can saturate at large score magnitudes, making gradients small. ReLU is $\max(0,z)$. For mutually exclusive classes, softmax gives $p_k = \exp(a_k) / \sum_j \exp(a_j)$ and cross-entropy for true class y is $-\log(p_y)$. Subtract the largest logit before exponentiation for numerical stability; probabilities are unchanged because the common factor cancels.

Stable softmax + cross-entropy

$$u(k) = a(k) - \max_j a(j)$$

$$L = -\log(p(y))$$

$$p(k) = \frac{e^{u(k)}}{\sum_j e^{u(j)}}$$

Each $p(k)$ lies between 0 and 1; all $p(k)$ values sum to 1.

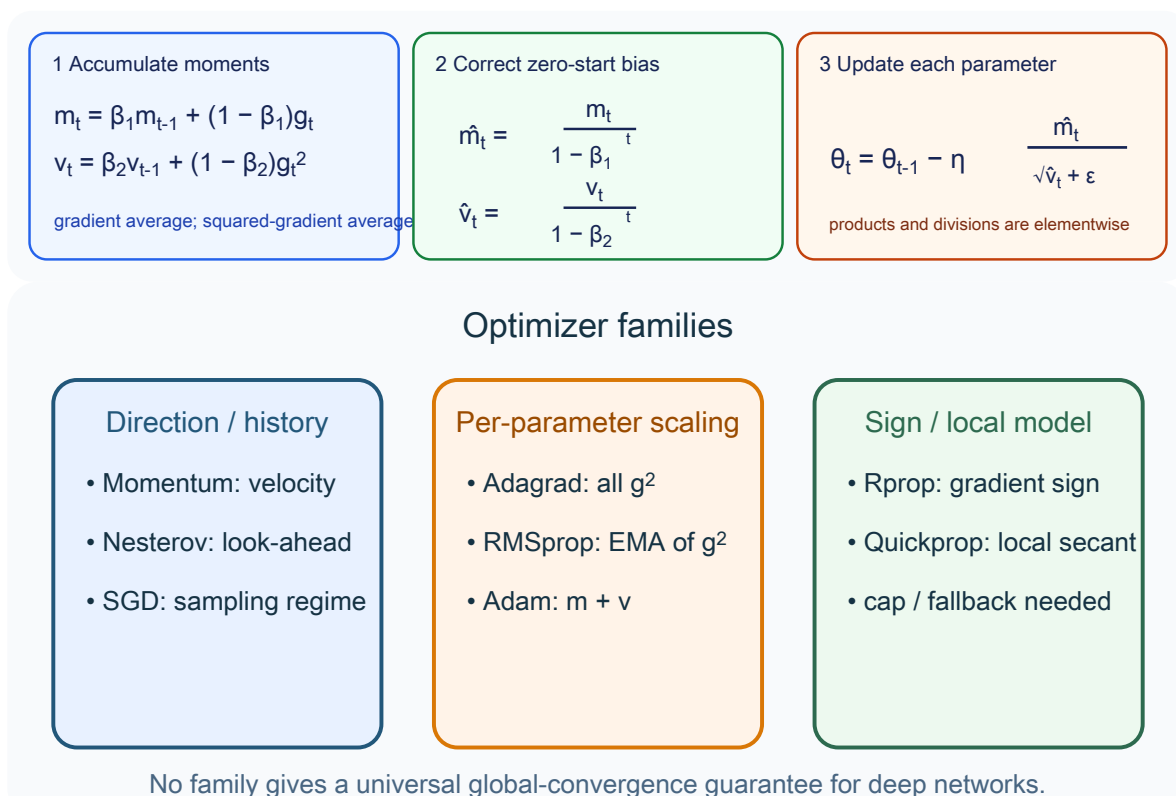
Worked example. With logits (2,1,0), subtract 2 to get (0,-1,-2). Using approximate exponentials (1,0.368,0.135), the sum is 1.503, so probabilities are approximately (0.665,0.245,0.090).

Checked sigmoid example. For $x=(2,-1)$, a hidden ReLU unit with weights (0.5,-1), bias 0.5 gives score and activation 2.5. An output weight 0.8 and bias -1 give logit 1, so sigmoid probability $1/(1+\exp(-1))$ is approximately 0.731.

Session 4 - Optimization of deep models

Loss is nonconvex when its landscape is not one bowl; it can contain local minima, saddles that rise in some directions and fall in others, and nearly flat plateaus. Batch gradient descent (GD) uses all examples, stochastic GD (SGD) one example, and minibatch SGD a subset. Momentum accumulates direction; Nesterov evaluates a gradient after a look-ahead. Rprop adapts a separate step from each gradient sign. Quickprop uses a safeguarded local quadratic/secant estimate. Adagrad accumulates all squared gradients; RMSprop exponentially discounts older squares.

At step t , θ_t is the parameter vector, g_t its loss gradient, m_t an exponential average of gradients, and v_t an exponential average of squared gradients. The learning rate is $\eta > 0$; β_1, β_2 are decay factors between 0 and 1; $\epsilon > 0$ prevents division by zero. Hats mark bias-corrected moments. All powers, roots, and divisions in the Adam equations are elementwise. Bias correction offsets zero initialization early. No optimizer here guarantees a deep network's global optimum.



Worked example. With $g_1=-3$, zero moments, $\beta_1=0.8$, $\beta_2=0.9$, and $\eta=0.02$, $m_1=-0.6$, $v_1=0.9$, corrected moments are -3 and 9, so the first parameter change is about +0.02 when epsilon is negligible.

A learning rate that is too large can make loss oscillate or diverge; one that is too small can make improvement stall. Compare loss curves and update magnitudes before changing the optimizer.

Session 5 - Regularization for deep models

Training data fits parameters; validation data selects settings; the test set is held for final evaluation. High train and validation errors suggest underfitting. Low train error but worse/rising validation error suggests overfitting.

The parameter vector θ contains learned weights and biases. The regularization strength $\lambda \geq 0$ controls the added penalty: L1 adds $\lambda \sum |\theta_i|$; L2 adds $\lambda \sum \theta_i^2$. Dropout randomly masks training activations and uses the corresponding deterministic inference convention. A Jacobian is a table of output derivatives with respect to inputs; multiplying many small or large derivative/Jacobian factors can make gradients vanish or explode. Gradient clipping caps a gradient's norm or entries before an update and can limit an explosion, but it does not repair its underlying cause. Initialization aims to preserve signal scale. Batch normalization uses minibatch statistics in training and stored estimates at inference.

L1 penalty

$$\lambda \sum_i |\theta_i|$$

can encourage exact zeros

L2 penalty

$$\lambda \sum_i \theta_i^2$$

smoothly discourages large values

$\lambda \geq 0$; penalty choice changes the training objective, not a guaranteed test result.

Training

$$\tilde{h} = \frac{m \odot h}{q}$$

$$m_i \in \{0, 1\}, P(m_i=1)=q$$

Inference

$$\tilde{h} = h$$

no random mask

Expected training activation is preserved when $0 < q \leq 1$.

Batch normalization

activation x

normalize
batch μ, σ^2 in training

scale and shift
 $y = \gamma \hat{x} + \beta$

Inference uses stored population estimates, not the current test batch.

BN does not universally remove dataset shift.

Xavier / Glorot

$$\text{Var}(W_{ij}) \approx \frac{2}{\text{fan-in} + \text{fan-out}}$$

He / rectifier

$$\text{Var}(W_{ij}) \approx \frac{2}{\text{fan-in}}$$

Distribution details and activation assumptions must be stated.

The BN paper historically described reducing internal covariate shift, meaning a change in a layer's input distribution as earlier layers update. That motivation is not a universal claim that BN eliminates dataset shift, which is a distribution difference between training and deployment or other data settings.

Worked example. If data loss is 0.30, $\lambda=0.04$, and weights are (1,-2,2), the L2 objective is $0.30+0.04(1+4+4)=0.66$.

Sessions 6-7 - Convolutional networks

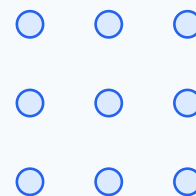
This package uses cross-correlation: slide a small number grid called a kernel without flipping it, multiply overlapping entries, and add. With input size N , padding P , kernel size K , stride S , and dilation D , effective kernel size is $K_{\text{eff}}=D(K-1)+1$ and each output spatial size is $\text{floor}((N+2P-K_{\text{eff}})/S)+1$. C_{in} and C_{out} are input/output channel counts; a standard $K_h \times K_w$ layer has $K_h \times K_w \times C_{\text{in}} \times C_{\text{out}} + C_{\text{out}}$ parameters. Max pooling keeps the largest local value; average pooling keeps the local mean. Neither learns a kernel.

Dilated cross-correlation

$$K_{\text{eff}} = D(K - 1) + 1$$

$$N_{\text{out}} = \text{floor} \left(\frac{N + 2P - K_{\text{eff}}}{S} \right) + 1$$

$$\text{parameters} = K_h K_w C_{\text{in}} C_{\text{out}} + C_{\text{out}}$$



3 × 3 taps, dilation $D=2$

Apply the output formula separately to height and width.

One cross-correlation position

$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \rightarrow$$

$$1(1) + 2(-1) + 3(0) + 1(2) = 1$$

Kernel is not flipped under the stated deep-learning convention.

Receptive field is the input region influencing one activation. Deep stacks enlarge it. A residual block learns a change $F(x)$ to input x and outputs $F(x)+x$; addition requires matching shapes or a projection. AlexNet is relevant as a landmark trained CNN and ResNet for residual learning, not as memorization trivia.

Transfer learning reuses source features. Freeze when target data are scarce and representations transfer well; fine-tune when adaptation is worth the data/compute/overfit risk. Domain mismatch can make reuse ineffective.

Worked example. For $N=18, K=3, P=1, S=2, D=2, K_{\text{eff}}=5$ and output size is $\text{floor}((18+2-5)/2)+1=8$. With $C_{\text{in}}=3, C_{\text{out}}=6$, parameter count is $3*3*3*6+6=168$.

Session 8 - Integrated review

No new theory is introduced. Diagnose by evidence: representation failure (single perceptron on XOR), shape failure (matrix/channel mismatch), optimization failure (flat or unstable training), generalization failure (train-validation gap), or design mismatch (invalid convolution/residual/transfer choice). State one governing principle, compute or trace it, state conditions, then interpret the result and propose one targeted check.

Mixed worked checks

- 1 Perceptron: $x=(2,1)$, $w=(-0.5,1)$, $b=0.2$ gives $z=0.2$, hence 1 under the fixed threshold.
- 2 Chain rule: if $h=3x+2$, $L=h^2$, then at $x=1$, $h=5$ and $dL/dx=2h*3=30$.
- 3 CNN: $12 \times 12 \times 4$, two 5×5 kernels, no padding, stride 1, dilation 1 gives $8 \times 8 \times 2$ and $5*5*4*2+2=202$ parameters.
- 4 Diagnosis: falling training loss with rising validation loss is an overfitting pattern; inspect the validation minimum and regularization before touching the test set.

Conceptual Drills - Answer Free

Answer each in 3-6 sentences. State every condition behind a property; do not use absolute claims where the result is conditional.

Sessions 1-2

C01

Separate AI, ML, and DL, and give one valid system that shows why the nesting is not equality.

C02

Compare supervised, unsupervised, semi-supervised, and reinforcement learning by the information available during learning.

C03

Explain the limited biological-neuron analogy, naming dendrites, axon hillock, axon, and the corresponding artificial-neuron roles.

C04

Under the rule $z \geq 0 \rightarrow 1$, explain why a step activation is nonlinear while one perceptron's decision boundary remains linear.

C05

State all conditions required by the classical perceptron convergence guarantee and explain why XOR violates the relevant condition.

C06

Distinguish depth, width, representation capacity, trainability, and generalization. Include the limited meaning of universal approximation.

Session 3

C07

Separate the roles of forward propagation, backpropagation, and an optimizer.

C08

Compare sigmoid and softmax outputs and identify a task structure suitable for each.

C09

Explain why subtracting a common maximum logit improves numerical stability without changing softmax probabilities.

Session 4

C10

Compare batch GD, SGD, and minibatch SGD by gradient data, noise, and computational trade-offs.

C11

Distinguish a saddle point, plateau, and local minimum using curvature or directional behaviour.

C12

Organize Rprop, Quickprop, momentum, Nesterov, Adagrad, RMSprop, and Adam into families by the information each uses. State one safeguard or limitation for each family.

Session 5

C13

Explain the separate jobs of training, validation, and test sets and why repeated test-set consultation is invalid model selection.

C14

Compare L1, L2, and dropout by mechanism, train/inference behaviour, and likely effect.

C15

Explain vanishing/exploding gradients, initialization, and batch normalization without claiming that BN universally solves dataset shift.

Sessions 6-7

C16

Distinguish mathematical convolution from the cross-correlation convention used in CNNs.

C17

Explain how padding, stride, dilation, and pooling each change a feature map.

C18

Explain why convolutional parameter count does not grow with input image height and width for a fixed kernel/channel design.

C19

Describe how learned kernels and receptive fields support hierarchical image features.

C20

Explain a residual connection, its shape condition, and why it helps without guaranteeing successful training.

C21

Compare freezing and fine-tuning for transfer learning, including data size, compute, overfitting, and domain mismatch.

Session 8

C22

A model cannot learn XOR with one perceptron. Diagnose the failure and name a bounded architectural correction.

C23

A deep sigmoid network has nearly zero gradients in early layers while training and validation losses remain high. Give two plausible mechanisms, two checks, and one targeted response.

C24

A pretrained natural-image CNN performs poorly on medical scans. Build a diagnosis that separates underfitting, overfitting, and source-target domain mismatch.

Numerical and Applied Drills - Answer Free

For every calculation-bearing response, show the governing formula, substitution, non-obvious steps, units/shapes/conditions, and interpretation.

Sessions 1-2

N01

Use the fixed step convention on $x=(2,-1)$, $w=(0.6,0.8)$, $b=-0.1$. Compute z and the prediction.

N02

Start with $w=(0.2,-0.4)$, $b=0.1$, $\eta=0.25$. For $x=(3,2)$, target $y=1$, prediction $\hat{y}=0$, compute the updated weights and bias.

N03

With $w=(1,1)$ and the fixed threshold, derive the complete interval of biases implementing AND; test both endpoints.

N04

For $w=(-1,-1)$, find the complete bias interval that implements NOR under the fixed threshold and verify all rows.

N05

Count weights, biases, and total parameters in a dense $7 \rightarrow 4 \rightarrow 3$ network. State the shape of each weight matrix and bias vector.

N06

Two affine layers have scalar rules $h=4x-3$, $y=-2h+5$. Collapse them to one affine rule and evaluate it at $x=-1$.

Session 3

N07

For $x=2$, compute the forward values and dL/dx when $z=3x-4$, $h=\text{ReLU}(z)$, and $L=(h-5)^2$.

N08

Compute stable softmax probabilities to three decimals for logits $(4,2,1)$, then compute cross-entropy when the true class is the second class.

N09

A layer receives a 6×1 column and produces 5×1 ; the next produces 2×1 . State all weight/bias shapes and total parameters.

Session 4

N10

For Adam at $t=1$, use $g_1=4$, zero moments, $\beta_1=0.9$, $\beta_2=0.999$, $\eta=0.005$, and negligible epsilon. Compute all moments, corrected moments, and the parameter change.

N11

Adagrad starts with squared-gradient accumulator $G_0=0$. For gradients $2,-1,3$, compute G_1, G_2, G_3 and effective scalar multipliers $\eta/\sqrt{G_t}$ for $\eta=0.12$ and negligible epsilon.

N12

An RMSprop accumulator obeys $v_t=0.8v_{t-1}+0.2g_t^2$, with $v_0=1$. For $g_1=3$, compute v_1 and the scaled step magnitude $0.02*|g_1|/\sqrt{v_1}$.

Session 5

N13

Data loss is 0.55 , $\lambda=0.03$, and weights are $(-2,4,-1)$. Compute L1- and L2-regularized objectives under the definitions in the guide.

N14

In inverted dropout, $h=(3,6,9,12)$, keep probability $q=0.75$, and mask $m=(1,0,1,1)$. Compute the training output and state the inference output.

N15

Training loss over four checks is $(0.62, 0.38, 0.22, 0.14)$ and validation loss is $(0.65, 0.41, 0.36, 0.49)$. Identify the best checkpoint under validation selection, quantify the final gap, and interpret.

Sessions 6-7

N16

Input is $28 \times 28 \times 3$; a layer has 8 kernels of 5×5 , padding 2, stride 2, dilation 1. Compute output shape and parameters.

N17

Input is $21 \times 17 \times 6$; a layer has 10 kernels of 3×4 , padding $(2,1)$, stride $(2,1)$, dilation $(2,3)$. Compute both effective kernel sizes, output shape, and parameters.

N18

On a 4×4 single-channel patch with rows $(1,2,0,1)$, $(3,1,2,2)$, $(0,1,3,1)$, $(2,0,1,4)$, apply the 2×2 cross-correlation kernel with rows $(1,-1)$, $(0,2)$, no padding and stride 1. Compute the full output.

N19

Starting with receptive field $r_0=1$ and jump $j_0=1$, compute r_j after a 3×3 stride-2 layer, then a 3×3 stride-1 layer, both dilation 1.

N20

A residual main branch outputs $14 \times 14 \times 32$, while the shortcut is $28 \times 28 \times 16$. Identify both shape mismatches and specify the required projection stride/output channels.

N21

You have 900 labelled target images and a source model trained on a closely related 2-million-image dataset. Propose freeze/fine-tune stages, validation checks, and one stop condition; no numerical calculation is required.

Session 8

N22

A $12 \rightarrow 10 \rightarrow 4$ classifier has low training and validation accuracy. Add one hidden layer of width 6: compute old/new parameter counts and interpret whether the change alone proves the diagnosis.

N23

For a $32 \times 32 \times 3$ input, compare parameters of a dense layer with 20 outputs against a 3×3 convolution with 20 output channels and biases. Compute the ratio and interpret parameter sharing.

N24

A model's deployment accuracy falls from 88% on the held-out test distribution to 61% after a camera change, while the old test score remains 88%. Compute the percentage-point change and give a conditional diagnosis and validation plan.

Source-Aware BITS-Style Practice - Answer Free

All five prompts are newly authored practice for the current handout scope. The inaccessible third-party previews in the research ledger are not reproducible question evidence and are not used to claim historical origin.

H01 - MLP decision boundary

Classification: Authored BITS-style; institution/source/term/year: not applicable (newly authored practice); course: target AMLCSZG511, Deep Neural Networks. Third-party hosting: not applicable.

A binary MLP uses only affine transformations between input and output, with no nonlinear hidden activation. Determine the geometric type of decision boundary it can produce and justify the conclusion algebraically.

H02 - Training/validation diagnosis

Classification: Authored BITS-style; institution/source/term/year: not applicable (newly authored practice); course: target AMLCSZG511, Deep Neural Networks. Third-party hosting: not applicable.

Training loss continues falling while validation loss rises for five successive checkpoints. Diagnose the pattern, identify the checkpoint-selection rule, and propose two bounded responses without consulting the test set.

H03 - CNN shape and parameters

Classification: Authored BITS-style; institution/source/term/year: not applicable (newly authored practice); course: target AMLCSZG511, Deep Neural Networks. Third-party hosting: not applicable.

An input of shape 30 x 30 x 3 enters 12 kernels of size 3 x 3, padding 1, stride 2, dilation 1, with one bias per output channel. Compute output shape and trainable parameters, showing conditions and interpretation.

H04 - NOR perceptron

Classification: Authored BITS-style; institution/source/term/year: not applicable (newly authored practice); course: target AMLCSZG511, Deep Neural Networks. Third-party hosting: not applicable.

Under $z \geq 0 \rightarrow 1$, construct a two-input NOR perceptron, derive the valid bias interval for your chosen weights, and verify all truth-table rows.

H05 - Fresh integrated practice

Classification: Authored BITS-style Institution: Birla Institute of Technology & Science, Pilani, WILP Course: target AMLCSZG511, Deep Neural Networks Term/year/source: not applicable; newly authored from current handout scope Third-party hosting: not applicable.

A pretrained CNN works well on source validation but poorly on target validation. Distinguish domain mismatch from target overfitting, then propose a freeze/fine-tune experiment.